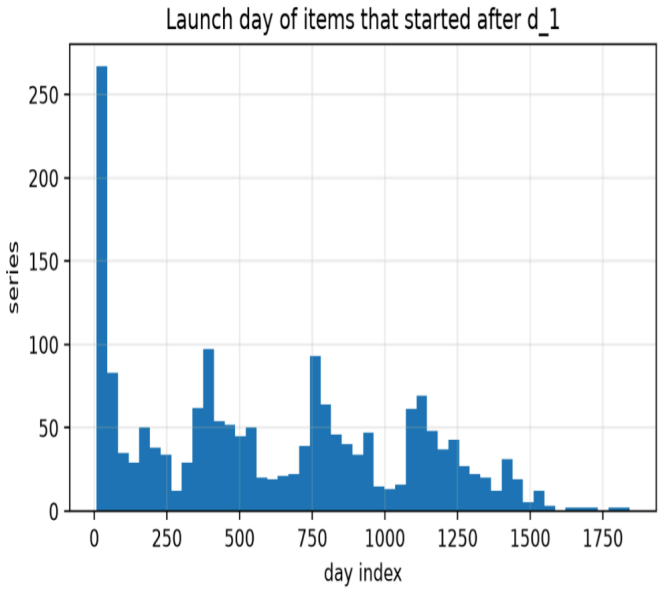
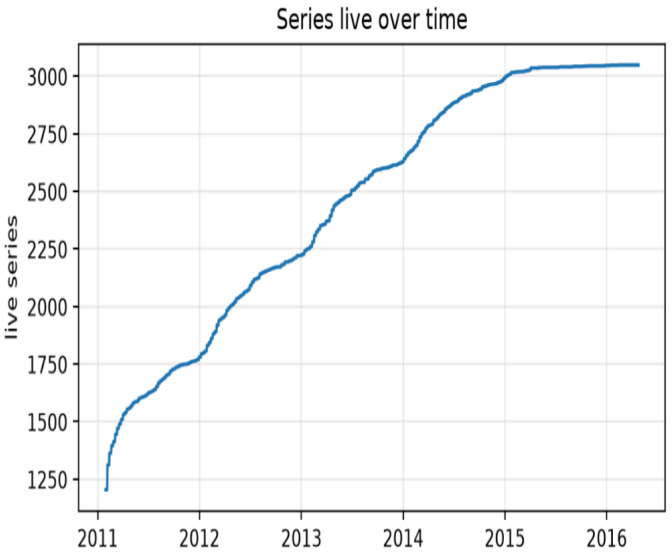
Subset `ca_1` — stores ['CA_1']: 3,049 series × 1,941 days. Training window: d_1 … d_1913 (2011-01-29 → 2016-04-24); held out: last 28 days.

Launches and Lifecycle:

measure	series	share_%
Live since d_1 (left-censored)	**1204**	**39.5**
Launched inside training window	**1845**	**60.5**
Not live in training window	**0**	**0.0**
Short history (< 56 live days)	**0**	**0.0**
No sales at all in training window	**0**	**0.0**
Trailing zeros ≥ 28 days	**185**	**6.1**
Trailing zeros ≥ 56 days	**129**	**4.2**
Longest zero run while live ≥ 28 days	**2425**	**79.5**

Sales on days with no price entry: **0** (expected 0; if not, `first_live_d` is not a reliable on-sale marker).





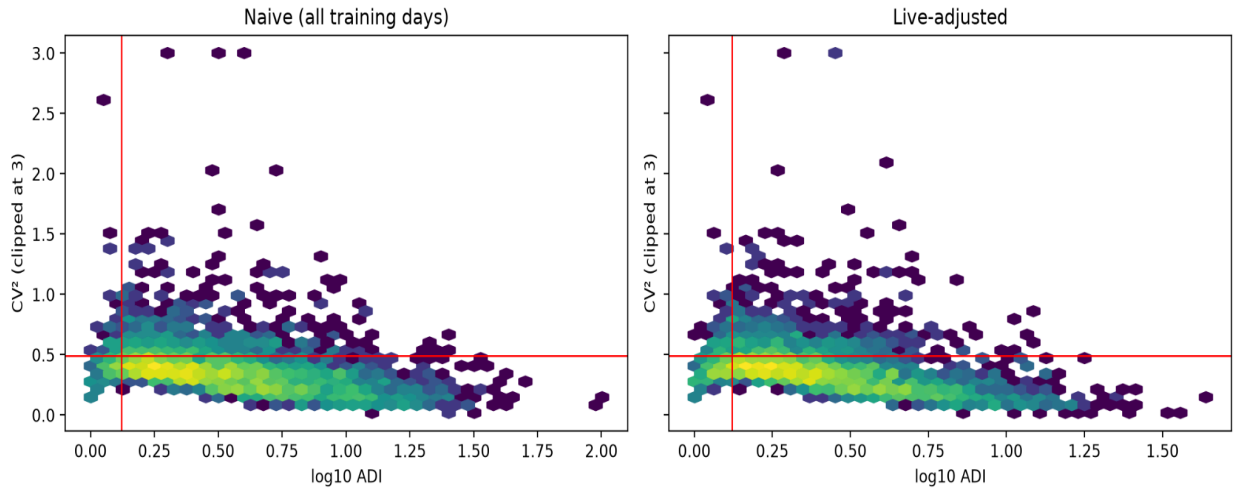
Intermittency: naive vs. live-adjusted:

4.3% of series change class when the pre-launch zeros are removed. Rows: naive class; columns: live-adjusted class.

naive	Smooth	Erratic	Intermittent	Lumpy
Smooth	133	0	0	0
Erratic	0	44	0	0
Intermittent	98	0	2,197	0
Lumpy	0	33	0	544

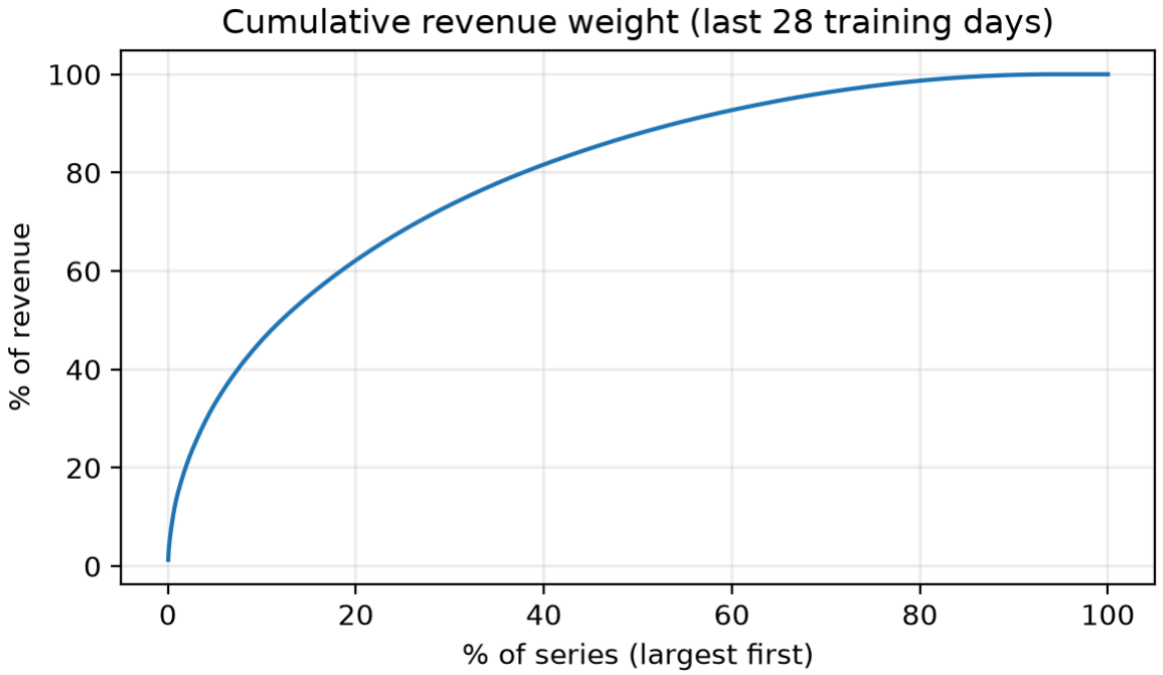
Live-adjusted class mix by category (% of series)

cat_id	Intermittent	Erratic	Lumpy	Smooth
FOODS	58.5	4	27.1	10.4
HOBBIES	79.3	1.6	18.1	1.1
HOUSEHOLD	86.8	1	5	7.3



Series needed to reach 50 / 80 / 90 / 99% of revenue weight: **368** (50%), **1,151** (80%), **1,646** (90%), **2,495** (99%)

class	series	series_%	units_%	revenue_weight_%
Smooth	231	7.6	29.7	24.7
Erratic	77	2.5	8.3	4.2
Intermittent	2197	72.1	41.9	59.5
Lumpy	544	17.8	20.1	11.6



Is the subset representative?

store	in_subset	units_share_%	FOODS_%	HOBBIES_%	HOUSEHOLD_%	weekday_profile_corr	level_28d_corr
CA_1	TRUE	11.7	69.9	11.4	18.7	1	1
CA_2	FALSE	8.7	61.2	11.2	27.6	0.984	0.558
CA_3	FALSE	17	67.2	8.6	24.2	0.975	0.954

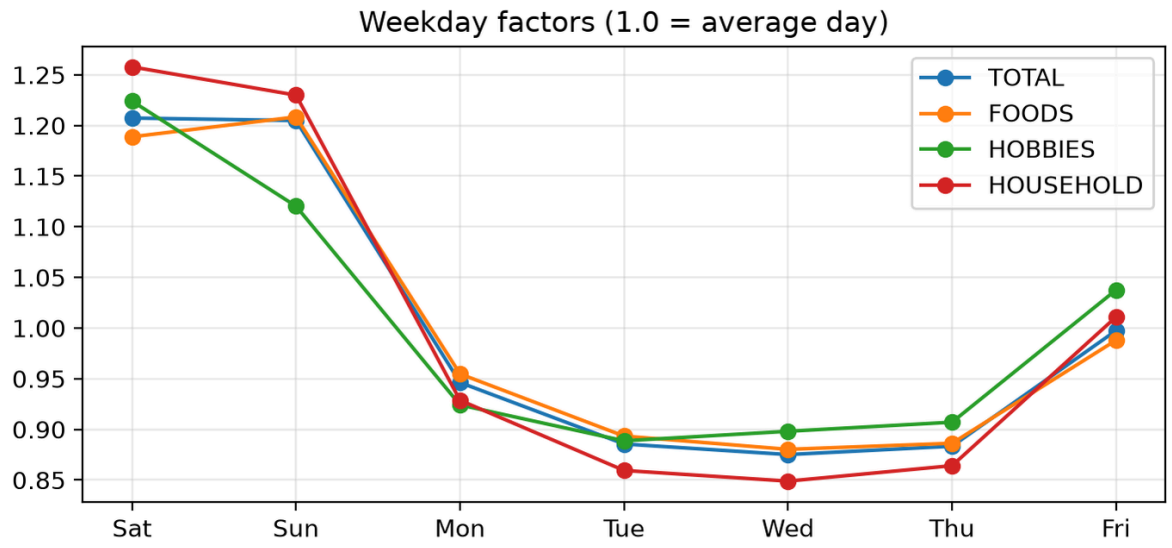
CA_4	FALSE	6.2	68.7	13.8	17.5	0.98	0.955
TX_1	FALSE	8.5	67.5	7.7	24.8	0.985	0.916
TX_2	FALSE	11	69.5	8.8	21.7	0.997	0.657
TX_3	FALSE	9.3	68.4	8.7	23	0.992	0.88
WI_1	FALSE	7.8	66.8	12.7	20.5	0.921	0.785
WI_2	FALSE	10	72.9	5.7	21.5	0.894	0.817
WI_3	FALSE	9.8	73.6	7.3	19.1	0.971	-0.09

Zero share per series (naive, full history incl. holdout; used only to compare the subset with the full portfolio):

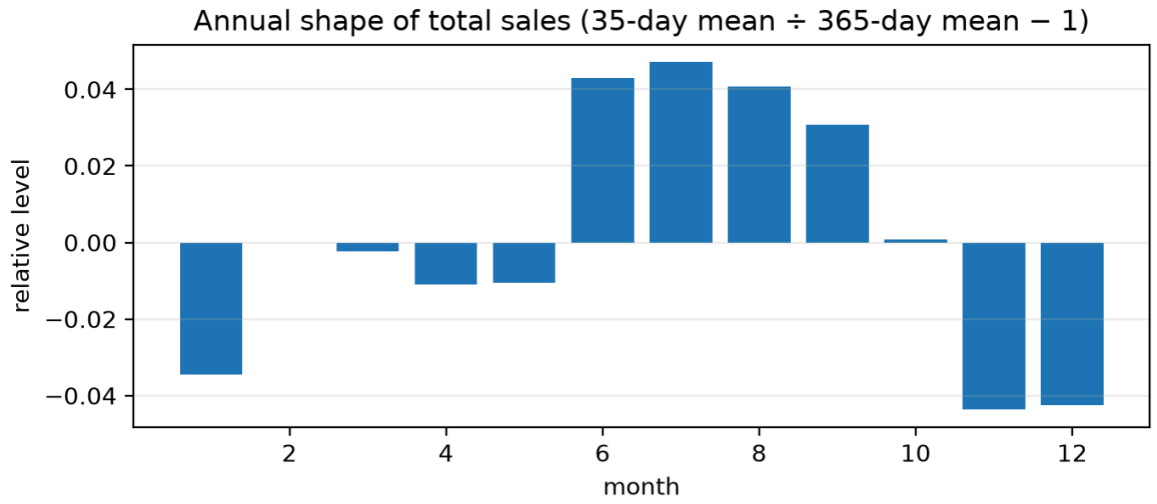
quantile	zero_share_subset	zero_share_all
0.1	0.309	0.344
0.25	0.478	0.535
0.5	0.681	0.733
0.75	0.823	0.865
0.9	0.903	0.928

Calendar effects against a trend baseline:

weekday	TOTAL	FOODS	HOBBIES	HOUSEHOLD
Saturday	1.207	1.189	1.224	1.258
Sunday	1.205	1.208	1.12	1.23
Monday	0.946	0.955	0.924	0.929
Tuesday	0.886	0.893	0.889	0.859
Wednesday	0.875	0.88	0.898	0.849
Thursday	0.883	0.886	0.907	0.864
Friday	0.998	0.988	1.038	1.011



Yearly rows: first and last years are partial; compare complete years only. `weekly_seasonal_strength` = 1 − var(residual)/var(ratio): close to 1 means the weekday pattern dominates short-term variation.



Named events (30 distinct, 154 event days). Percent difference vs. the weekday-and-trend baseline, all stores:

event	type	n	total_%	FOODS_%	HOBBIES_%	HOUSEHOLD_%
Christmas	National	5	-100	-99.9	-100	-100
Thanksgiving	National	5	-27.7	-21.8	-52.8	-35.4

LaborDay	National	5	25.1	27.4	6.5	24.9
NewYear	National	5	-21.8	-23.2	-26.9	-14.8
Mother's day	Cultural	5	-12.1	-9.7	-12.4	-19.6
Halloween	Cultural	5	-11.2	-12.1	-5.5	-11.1
Easter	Cultural	6	-11.1	-8.1	-20	-16.9
Father's day	Cultural	5	-11.1	-10.2	-9.2	-14.8
VeteransDay	National	5	8	11	3.6	0.4
ColumbusDay	National	5	6.9	9.9	0.5	0.1
ValentinesDay	Cultural	6	-6.5	-3.6	-9.5	-14.2
Cinco De Mayo	Cultural	5	5.2	7.6	0.7	-0.2
NBAFinalsStart	Sporting	5	5	5.3	-2.1	7.1
Chanukah End	Religious	5	3.9	4.7	2.1	3.4
Presidents Day	National	6	3.7	1.5	0.3	12.7
NBAFinalsEnd	Sporting	5	-3.6	-2.7	-5.6	-5
Eid al-Fitr	Religious	5	-3.5	-4.2	-2.6	-1.4
StPatricksDay	Cultural	6	-3.2	-3.5	-1.8	-2
OrthodoxChristmas	Religious	5	3	6.3	-5.1	-4.3
OrthodoxEaster	Religious	5	-2.8	-0.7	-7.4	-7
MartinLutherKingDay	National	5	2.8	0.9	4.3	8.7
MemorialDay	National	5	2.5	0.4	-0.1	11

LentStart	Religious	6	2.2	2.7	4.2	-0.6
IndependenceDay	National	5	-1.5	3.8	-18.4	-11.4
LentWeek2	Religious	6	-1.4	-3	2.9	1.9
Ramadan starts	Religious	5	1.3	0.7	4.2	2
Pesach End	Religious	5	-1	-3.5	6.4	3.8
Purim End	Religious	6	0.8	1	2.6	-0.1
SuperBowl	Sporting	6	0.5	4.9	-8.2	-10.2
EidAlAdha	Religious	5	0.3	1.2	-0.6	-1.9

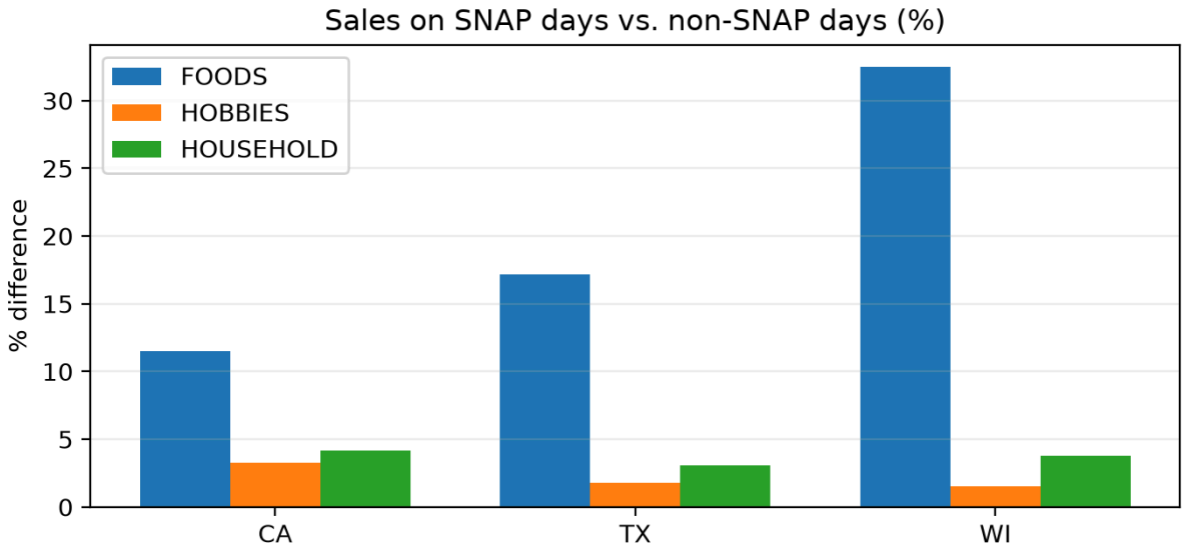
Dec 25 closure — near-zero total sales across all stores:

date	total_units	ratio_to_baseline_%
2011-12-25	13	0
2012-12-25	11	0
2013-12-25	20	0.1
2014-12-25	20	0.1
2015-12-25	14	0

SNAP effect by state × category (all stores in the state):

state	category	snap_days	snap_effect_%
CA	FOODS	630	11.5
CA	HOBBIES	630	3.3
CA	HOUSEHOLD	630	4.2
TX	FOODS	630	17.2

TX	HOBBIES	630	1.8
TX	HOUSEHOLD	630	3.1
WI	FOODS	630	32.5
WI	HOBBIES	630	1.5
WI	HOUSEHOLD	630	3.8



Price effects

measure	value
Complete weeks analysed	273
Live series-weeks	670,971
Week-to-week price changes (% of consecutive live pairs)	0.94
Median |price change| when it changes (%)	7.4
Series with ≥ 1 price change (%)	71.1
Live series-weeks priced > 5% below series median (%)	8.1

Bootstrap resamples (0 = point estimate only) = 40

group	series	slope_log1p_units_per_log_price	ci95_low	ci95_high
all	2169	-1.538	-1.756	-1.281
dense (≤10% zero weeks)	706	-1.058	-1.348	-0.758
FOODS	1123	-2.339	-2.601	-2.006
HOBBIES	377	-0.732	-1.107	-0.361
HOUSEHOLD	669	-0.742	-1.166	-0.456

Negative = demand falls when the series' own price rises. Slopes on `log1p(units)` are attenuated for sparse series; the *dense* row is the cleanest read.

Known-future price coverage. Of the 3,049 series live on the last training day, share with a price in each week after it (validation + competition horizon). Prices can be supplied to Timestral as known-future covariates only where this is ≈ 1.

wm_yr_wk	share_with_price
11613	1
11614	1
11615	1
11616	1
11617	1
11618	1
11619	1
11620	1
11621	1

STUMPY readiness: constant windows

class	series	median_const_share_m7	median_const_share_m28	series_const_gt50%_m7	series_const_gt50%_m28
Intermittent	2197	0.2	0.078	195	25
Lumpy	544	0.172	0.084	36	6
Smooth	231	0.015	0	0	0
Erratic	77	0.037	0.003	0	0

Starting-point eligibility for a z-normalised profile (≥ 365 live days and ≤ 50% constant windows): **2,806** series for m = 7, **3,006** for m = 28, out of 3,049. The rest need smoothed/aggregated input, `normalize=False`, or an annotation vector.

Phase 1 checkpoint

Numbers from this run (subset of 3,049 series, training window only):

- **Lifecycle: 39.5% live since d_1; 0.0% short history; 0.0% no sales; 6.1% end with ≥ 28 zero days.**
- **Intermittency: 4.3% of series change SBC class once pre-launch zeros are removed.**
- **STUMPY: 92.0% eligible for z-normalised m = 7; 98.6% for m = 28.**

What the results establish

- Weekly pattern: it is strong. Saturday and Sunday run about 1.2× the average, and Tuesday to Thursday about 0.88×, in every category. `m = 7` is clearly justified.
- Data quality: `first_live_d` is reliable, with 0 sales on days with no price. Prices exist for every series in all 9 future weeks.
- Revenue concentration: 368 series (12%) carry half the revenue, and 1,151 (38%) carry 80%.
- Assortment growth: live series grow from 1,204 to 3,049 over the period. Part of any aggregate growth is new items, not demand.
- SBC classes: the class boundary is soft. The densest cluster sits at CV^2 of about 0.4, just under the 0.49 line, so Intermittent vs. Lumpy labels are unstable. Stratify on volume and ADI, not CV^2 alone.

Revised Conclusions

1. Intermittent and lumpy series are not low-volume. They are 90% of series, 62% of units and 71% of revenue weight (59.5% intermittent, 11.6% lumpy). They are the bulk of the evaluation weight, so they need real STUMPY treatment, not a fallback.
2. My readiness flag was too weak. It only checks for constant windows. STUMPY has explicit handling for those (`T_subseq_isconstant`), so they don't produce NaNs, but all-zero windows still match each other and dominate motif lists. A window with a single sale is also a worse problem: z-normalization throws away the size of the sale and keeps only its position. So "98.6% eligible" overstates readiness. I'll add a non-zero-density criterion to the notebook. I haven't confirmed STUMPY's exact convention for constant windows empirically, so I'll test it in Phase 2.
3. `m = 364` isn't well supported. The annual swing in the total is only about ±4%. At item level, 364-day windows need a similar year to compare against, and you only have about 5. Weekly aggregates with `m = 52`, on high-volume series, are a better test. `m = 28` is justified by the forecast horizon, not by months.
4. SNAP and events:
    - SNAP lifts vary by state, from +11.5% in CA to +32.5% in WI, and your subset is CA only.
    - Non-food categories also rise 1.5–4% on SNAP days. SNAP can't buy those items, so this is likely a day-of-month effect. The FOODS lift beyond that is roughly 8 points in CA, 15 in TX and 30 in WI.
    - Only about 8 events are large: Christmas (a closure), Thanksgiving, Labor Day, New Year, Mother's Day, Halloween, Easter and Father's Day. Each has n = 5 and no confidence intervals, so effects under about 5% are noise.
5. Elasticity: the notebook flagged the dense-series estimate (−1.06, CI −1.35 to −0.76) as the cleanest read, not −1.54 or −2.34. Only 0.9% of consecutive weekly pairs change price, so the signal is sparse. "Price will improve Timestral accuracy" is a hypothesis for the ablation, not a result.
6. "Subset is representative" is only partly true:
    - The weekday profile generalizes (correlation ≥ 0.89 across stores).
    - Level dynamics don't: the 28-day level correlation with CA_1 runs from −0.09 (WI_3) to 0.96, and is 0.56 even for CA_2.
    - Zero shares in CA_1 are 3–6 points lower than the full portfolio, so it is somewhat denser.
    - SNAP size differs about 3× by state.

The yearly and weekly-strength tables seem to be missing from the doc. If you have them, the yearly one would help separate demand growth from assortment growth.

Proposed next steps

1. Patch the Phase 1 notebook (small):
    - Add the density criterion.
    - Flag zero runs that are longer than a series' own sales rate would explain, to separate stockouts from sparse demand.
    - Add event windows of ±7 days.

    - Use non-food categories as a SNAP control.
    - Add a like-for-like trend using only series live since d_1.
2. Check generalization: run the same notebook on `WI_3` (the outlier) and `TX_1` (different state). If class mix and conclusions hold, scale up.
3. Run a Timestral baseline now. Zero-shot on `CA_1` (d_1914–d_1941) against seasonal-naive, scored with in-subset WRMSSE weights and pinball loss. It doesn't depend on STUMPY and tells us whether the Stumpy features have anything to improve.
4. Phase 2 STUMPY, in this order:
    - Aggregate levels (about 43 series) on daily `m = 7, 28` and weekly `m = 4, 13, 52`, with `stimp` for window choice. Check whether discords land near events.
    - Dense item-level series.
    - Intermittent and lumpy series, via weekly aggregates or `normalize=False` with annotation vectors.
    - `mpdist` clustering and FLUSS regimes.
5. Phase 3 features and ablation, using left-only matrix profile values to avoid leakage.

One note on tooling: `data_prep` currently rebuilds everything per run, so step 2 would repeat the full conversion. I'd add a `--subset-only` flag so new subsets take seconds.